

5 Interesting REFERENCE DESIGNS

Title of Reference Design	Backlight and Smart Lighting Control	Remote Controlled BLDC Fan
Image		
Key Attraction	The reference design features an optical light sensor and a proximity sensor to detect ambient light intensity and human presence respectively. Based on the above factors, the design adjusts the brightness of a light that is most comfortable for the human eyes. ADI's solution can simultaneously transmit data and power beyond 1km.	This reference design is of a remote-controlled brushless DC (BLDC) motor driver for ceiling fan application. The BLDC fan offers higher efficiency and silent operation compared to a conventional fan.
Highlights	The reference design features an optical light sensor which detects the intensity of light in the surrounding environment. The design also has a proximity sensor which detects human presence and adjusts the brightness of a light or a display that is most comfortable for the human eyes. A simple algorithm included in the reference design package determines the ideal backlight brightness when the backlight is activated. Proximity wake-up is enabled by Texas Instruments' capacitive-to-digital converter. A capacitive sensor located around the edge of the board provides a 360-degree sensing field around the board. The reference design enables more comfortable viewing experience and higher energy saving for the customers.	The reference design is for a brushless DC (BLDC) ceiling fan which can be controlled using most off-the-shelf IR remotes. It has an efficiency of over 85% and consumes a low power of 30-40 watts, which is less than half the power consumed by existing fans. The reference design can work over a wide range of input voltage ranging from 90V AC to 265V AC. Unlike the existing induction motors used in conventional fans, the BLDC motors do not have brushes, so they last longer. Furthermore, this reference design with BLDC motors provides a higher torque/power-to-weight ratio and runs efficiently and silently compared to brushed DC motors. The reference design employs a 24V three-phase motor driver to drive motors with sinusoidal current and sensorless control. The reference design also features a value-line processor that enables remote control by decoding infrared signals. The included firmware allows easy integration with standard infrared remotes based on the NEC transmission protocol, thus allowing the users to use most off-shelf remotes to control the fan.
Application	The module can control the light of a room or the back-light of a display.	The reference design is suitable for use in residential and commercial purposes.
OEM Brand	Texas Instruments	Texas Instruments
URL	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-backlight-and-smart-lighting-control	https://www.electronicsforu.com/electronics-projects/industry-grade-projects/reference-design-for-a-remote-controlled-bldc-fan
QR Code		